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The output of comparator A1 goes high when the voltage at its non-inverting input is greater than the voltage at the inverting input. Comparators A2 and A3 act as inverting and non-inverting buffers, respectively. Resistors R4 and R5 form a voltage divider which provides reference voltage at pins 7 and 8 for comparators A2 and A3, respectively.

Mode switch S2 is used to select the output of A2 (pin 1) or A3 (pin 14). Pole of switch S2 is connected to the base of transistor T1. The base of transistor T1 is driven into saturation via resistor R9, which is connected to unregulated 12V supply. Relay RL1 is connected to the collector of T1. Therefore T1 acts as a switch for relay RL1. Diode D5 across the coil of relay acts as a free-wheeling diode. The motor of the AC fan is connected to the circuit through relay contacts. Thus relay switches the fan on or off.

Temperature sensor IC LM335 acts as a zener diode. Its breakdown voltage is directly proportional to the absolute temperature at 10 mV/°K. Resistor R3 limits the current through IC4. Capacitor C4 bypasses any external noise.

Construction

An actual-size, single-side PCB for the automatic fan controller for ACs is shown in Fig. 3 and its component layout in Fig. 4. Assembling the circuit on a PCB minimises time and assembly errors.

Use bases for ICs LM3914 and IC LM339. Enclose the assembled circuit in a suitable cabinet. On the PCB, provide suitable connectors for switches S1 and S2 and potentiometer VR3 to extend these out from the cabinet through cable.

The sensor is brought out from the cabinet with a two-core cable. LED1 through LED10, switches S1 and S2, and potentiometer VR3 are mounted on the front panel of the cabinet. LED1 through LED10 are marked with calibrated temperature values.

Calibration. Calibrate the automatic temperature controller before putting it into the AC's circuit. Calibration is done for temperature control between 20°C and 29°C and temperature indication by LED1 through LED10.

First, with the help of a thermometer and digital voltmeter, set the temperature range to be indicated and controlled. Make sure that temperature sensor IC4 is free-standing in air, away from any source of heat, such as soldering iron or a heat-emitting lamp in the room.

Apply AC power to the circuit after finishing the construction. Do not connect motor to the unit yet. Using the thermometer, take the room temperature reading. Adjust preset VR2 such that the LEDs indicate a corresponding temperature. For instance, if the temperature is 24°C, LED1 through LED5 should glow, while LED6 through LED10 remain off.

Next, connect the digital voltmeter across potentiometer VR3 and adjust preset VR1 such that the voltage reading is exactly 0.111V. This sets the temperature range by potentiometer VR3 between 20°C and 29°C. ●

The author is a regular contributor to EFY